

A criticality-anchored radius cannot escape a slow saddle in bounded time

Abstract

A radius tied to a criticality measure inherits the measure's collapse near a saddle. We prove a lower bound on the escape time that uses no property of the model or of the subproblem solution beyond $\|s_k\| \leq \Delta_k$, so the stall survives perfect curvature information. For any cap $\bar{\mu}$ and any K there is an ε giving escape time above K . We then measure the constants. The fitted geometric rate agrees with the bound, the scaled escape time collapses onto a single value across three decades of $\bar{\mu}$ and five of ε , and the collapse constant is $\ln(1/(\sqrt{3}y_0))$ rather than $\ln(1/(2y_0))$.

1 The problem

For $\varepsilon > 0$ let

$$f_\varepsilon(x, y) = \frac{1}{2}x^2 + \varepsilon \left(\frac{1}{4}y^4 - \frac{1}{2}y^2 \right), \quad \nabla f_\varepsilon = (x, \varepsilon y(y^2 - 1)). \quad (1)$$

The critical points are the origin and $(0, \pm 1)$, with

$$\nabla^2 f_\varepsilon(0, 0) = \text{diag}(1, -\varepsilon), \quad \nabla^2 f_\varepsilon(0, \pm 1) = \text{diag}(1, 2\varepsilon). \quad (2)$$

The origin is a saddle, nonsingular for every $\varepsilon > 0$, so the second-order measure reports $\tau(0, 0) = \varepsilon$. The minimisers sit at distance 1 from the saddle whatever ε is. So ε tunes the escape rate without moving the geometry.

Lemma 1 (the line $\{x = 0\}$ is invariant). *Let $x_0 = 0$. Then $x_k = 0$ for every k , under truncated conjugate gradient and under the exact Moré-Sorensen solver alike.*

Proof. At $x = 0$ the gradient is $(0, \varepsilon y(y^2 - 1))$, which lies along e_2 , and the Hessian $\text{diag}(1, \varepsilon(3y^2 - 1))$ is diagonal, so $He_2 \parallel e_2$. Conjugate gradient starts at $s = 0$ with residual $-g \parallel e_2$ and first direction $-g$. If $s_j \parallel e_2$ and $d_j \parallel e_2$ then $Hd_j \parallel e_2$, the step length is a scalar, and s_{j+1} and d_{j+1} lie along e_2 . Induction gives the claim. For the exact solver, $(H + \sigma I)^{-1}$ is diagonal, so $s^* = -(H + \sigma I)^{-1}g \parallel e_2$. $\square$

Lemma 1 was also checked numerically. Over the whole grid of Section 3, $\max_k |x_k|$ is 0 in floating point, not merely small.

On the invariant line, for $|y| \leq 1$,

$$\|g\| = \varepsilon |y| (1 - y^2) \leq \varepsilon |y|. \quad (3)$$

The bound holds on $|y| \leq 1$, not only on $|y| \leq \frac{1}{2}$.

2 The proposition

We state the hypothesis on the radius rule rather than on a named rule, so that one statement covers both mechanisms.

Assumption 1 (criticality-anchored radius). There are $C > 0$ and $k_0 \geq 0$ such that

$$\Delta_k \leq C \max_{k_0 \leq j \leq k} \|g_j\| \quad \text{for all } k \geq k_0.$$

The running maximum is what makes the assumption cover R-DFO, whose radius is controlled by $\|g_{k-1}\|$ rather than by $\|g_k\|$. Nothing else in the proof needs it.

Proposition 2 (escape time). *Let $x_0 = 0$ and $0 < |y_{k_0}| \leq \frac{1}{2}$. Suppose the run satisfies Assumption 1 with constants C and k_0 . Write*

$$k_{\text{esc}} = \min\{k : |y_k| > \frac{1}{2}\}.$$

Then

$$k_{\text{esc}} > k_0 + \frac{\ln(1/(2|y_{k_0}|))}{\ln(1 + C\varepsilon)}. \quad (4)$$

Proof. Write $M_k = \max_{k_0 \leq j \leq k} |y_j|$. Every trust-region step satisfies $\|s_k\| \leq \Delta_k$, and by Lemma 1 the step lies along e_2 , so

$$|y_{k+1}| \leq |y_k| + \|s_k\| \leq |y_k| + \Delta_k.$$

The inequality also covers a rejected step, for which $y_{k+1} = y_k$.

Let $k_0 \leq k < k_{\text{esc}}$. Then $|y_j| \leq \frac{1}{2}$ for every $j \leq k$, so (3) applies at each such j and gives $\|g_j\| \leq \varepsilon |y_j| \leq \varepsilon M_k$. With Assumption 1,

$$\Delta_k \leq C \max_{k_0 \leq j \leq k} \|g_j\| \leq C\varepsilon M_k,$$

hence $|y_{k+1}| \leq M_k(1 + C\varepsilon)$. Since $1 + C\varepsilon > 1$,

$$M_{k+1} = \max\{M_k, |y_{k+1}|\} \leq M_k(1 + C\varepsilon),$$

and by induction $M_k \leq |y_{k_0}| (1 + C\varepsilon)^{k-k_0}$.

At $k = k_{\text{esc}}$ we have $|y_k| > \frac{1}{2}$, so $M_{k_{\text{esc}}} > \frac{1}{2}$ and $|y_{k_0}| (1 + C\varepsilon)^{k_{\text{esc}}-k_0} > \frac{1}{2}$. Taking logarithms gives (4). $\square$

Remark 1. The proof uses no property of the model or of the subproblem solution beyond $\|s_k\| \leq \Delta_k$. That is what makes it a statement about the radius axis. In particular it holds with the exact Hessian, and the experiments of Section 3 use the exact Hessian throughout.

Corollary 3 (an ε for every budget). *Fix $\bar{\mu} > 0$, $0 < |y_0| < \frac{1}{2}$ and $K > 0$. If*

$$0 < \varepsilon < \frac{\ln(1/(2|y_0|))}{\bar{\mu}K},$$

then any run satisfying Assumption 1 with $C = \bar{\mu}$ and $k_0 = 0$ has $k_{\text{esc}} > K$.

Proof. The hypothesis gives $\bar{\mu}\varepsilon < \ln(1/(2|y_0|))/K$, and $\ln(1+t) < t$ for $t > 0$, so $\ln(1 + \bar{\mu}\varepsilon) < \ln(1/(2|y_0|))/K$. Substituting into (4) gives $k_{\text{esc}} > K$. $\square$

2.1 The two mechanisms

R-grad($\omega, \bar{\mu}$) sets $\Delta_k = \mu_k \|g_k\|$ with $\mu_k \leq \bar{\mu}$, so Assumption 1 holds with $C = \bar{\mu}$ and $k_0 = 0$. This is immediate and needs no entry condition.

R-DFO(ω) is different, and the difference is not cosmetic. Its update is

$$\Delta_{k+1} = \begin{cases} \gamma_1 \Delta_k, & \rho_k < \eta_1, \\ \gamma_2 \Delta_k, & \rho_k \geq \eta_1 \text{ and } \Delta_k > \zeta \|g_k\|, \\ \gamma_3 \Delta_k, & \rho_k \geq \eta_1 \text{ and } \Delta_k \leq \zeta \|g_k\|, \end{cases} \quad (5)$$

so the radius is tied to the criticality measure only inside the band $\Delta_k \leq \zeta \|g_k\|$.

Proposition 4 (the R-DFO version, with its entry condition). *Suppose there is k_0 with $\Delta_{k_0} \leq \zeta \|g_{k_0}\|$. Then Assumption 1 holds for $k > k_0$ with $C = \gamma_3 \zeta$, and (4) applies with that constant.*

If no such k_0 exists, then $\Delta_k \rightarrow 0$ geometrically, $\|g_k\| \rightarrow 0$, and on the invariant line $|y_k| \rightarrow 0$. The run converges to the saddle and never escapes.

Proof. Write $S_k = \max_{k_0 \leq j \leq k} \|g_j\|$. At k_0 the hypothesis gives $\Delta_{k_0} \leq \zeta \|g_{k_0}\| \leq \gamma_3 \zeta S_{k_0}$, since $\gamma_3 > 1$. Suppose $\Delta_k \leq \gamma_3 \zeta S_k$. If $\Delta_k \leq \zeta \|g_k\|$, then (5) gives $\Delta_{k+1} \leq \gamma_3 \Delta_k \leq \gamma_3 \zeta \|g_k\| \leq \gamma_3 \zeta S_k$. Otherwise $\Delta_{k+1} \leq \max\{\gamma_1, \gamma_2\} \Delta_k < \Delta_k \leq \gamma_3 \zeta S_k$. Either way $\Delta_{k+1} \leq \gamma_3 \zeta S_k \leq \gamma_3 \zeta S_{k+1}$, which is Assumption 1 with $C = \gamma_3 \zeta$.

For the second statement, if $\Delta_k > \zeta \|g_k\|$ for every k , then both remaining branches of (5) contract, so $\Delta_k \leq \max\{\gamma_1, \gamma_2\}^k \Delta_0 \rightarrow 0$. Then $\|g_k\| < \Delta_k / \zeta \rightarrow 0$. On the invariant line $\|g_k\| = \varepsilon |y_k| (1 - y_k^2)$ with $1 - y_k^2 \geq \frac{3}{4}$ below the threshold, so $|y_k| \rightarrow 0$. $\square$

Remark 2. The entry condition bites. Started at $\Delta_0 = 1$ from $y_0 = 10^{-3}$ with $\varepsilon = 10^{-3}$, the run has $\Delta_0 / \|g_0\| \approx 10^6$, far outside the band, and it escapes at $k = 1$ for every ζ and every ε in the grid. Proposition 4 says nothing about such a run, and it should not. Section 3 therefore reports a separate arm started at $\Delta_0 = \zeta \|g_0\|$, which satisfies the entry condition at $k_0 = 0$ by construction.

2.2 Relation to Part II

Part II already contains the qualitative statement. Its `cond: s_k bounded by g_k`, labelled C.Sg, requires $\|s_k\| \leq \bar{\kappa} \|g_k\|$ near x^* from some index on. `thm: convergence to potential saddle point` shows that a subsequence converging to a nonsingular first-order critical point, together with C.Sg, forces the whole sequence to converge there. `cor: convergence criticality anchored saddle` then states that $\{x_k\}$ converges to x^* if the mechanism is R-grad with $\mu_k \leq \bar{\mu}$, or if it is R-DFO.

Proposition 2 is the quantitative complement. Part II says the criticality-anchored rules cannot leave the saddle asymptotically. This says how slowly they fail to leave it, and Corollary 3 converts that into a budget.

Two differences are worth recording. Part II's prose states that R-DFO does not satisfy C.Sg and reaches the same conclusion by another route. Assumption 1 is stated on Δ_k with a running maximum, and covers both rules for that reason. The induction of Proposition 4 is a relaxation of the bound inside the proof of `cor: convergence criticality anchored saddle`, which carries an additional γ_2^{k-k-1} factor and is therefore sharper.

Part II also states the repair that Section 3.4 measures. Near a saddle $\|g_k\| \rightarrow 0$ while $-\lambda_k^-$ stays bounded away from zero, so τ_k inherits a floor, and Part II records that it is `thm: second-order convergence`, not the mechanism, that converts the floor into escape.

3 Experiments

The grid takes ε over five decades from 10^{-1} to 10^{-5} , $\bar{\mu}$ and ζ over three decades, $y_0 \in \{10^{-3}, 10^{-6}\}$ and $x_0 \in \{0, 0.8\}$. The model is the true Hessian and the subproblem solver is truncated conjugate gradient throughout, so the stall appears with perfect curvature information. R-delta runs from the same starting points as the comparator.

Two tolerances are used and never pooled. The rate regime takes `tol` = 10^{-14} , which lets the dynamics run so that k_{esc} and the fitted rate are measurable. The outcome regime takes `tol` = 10^{-5} , which is where false convergence appears. A run that has not escaped within 50 000 iterations is reported `never` and is never treated as a number.

3.1 The radius is the binding quantity

Across the R-grad grid the median $\theta_k = \Delta_k / \|g_k\|$ equals $\bar{\mu}$ to six figures, and the trust-region constraint is active on every iteration of almost every cell. The radius is therefore what limits the step, rather than the subproblem solution.

3.2 The fitted rate

The escape phase is defined by $|x_k| \leq \varepsilon |y_k|$ and $|y_k| \leq \frac{1}{2}$. On the $x_0 = 0$ arm the first condition is vacuous and the phase is the whole pre-escape run. Regressing $\log |y_k|$ on k over that phase gives the fitted rate.

The fitted rate tracks $\ln(1 + \bar{\mu}\varepsilon)$ closely and from below, which is the side Proposition 2 predicts, since the proposition bounds growth from above. At $\bar{\mu}\varepsilon = 10^{-3}$ the fitted rate is 0.00099 against 0.00100; at 10^{-2} it is 0.00986 against 0.00995; at 10^{-1} it is 0.09454 against 0.09531. The one cell with $\bar{\mu}\varepsilon = 1$ gives 0.58752 against 0.69315, where the linearisation no longer applies. Figure 4 plots the comparison.

3.3 The collapse, and its constant

The scaled escape time $\bar{\mu}\varepsilon k_{\text{esc}}$ collapses onto one value across three decades of $\bar{\mu}$ and five of ε . The measured value is 6.362 at $y_0 = 10^{-3}$ and 13.273 at $y_0 = 10^{-6}$.

The predicted constant $\ln(1/(2y_0))$ is 6.215 and 13.122, leaving residuals of 0.147 and 0.151. The continuous limit of the recursion is $dy/dk = C\varepsilon y(1 - y^2)$, whose integral from y_0 to $\frac{1}{2}$ is

$$\int_{y_0}^{1/2} \frac{dy}{y(1 - y^2)} = \ln \frac{1}{\sqrt{3}y_0} + \mathcal{O}(y_0^2). \quad (6)$$

That gives 6.3585 and 13.2663, leaving residuals of 0.0035 and 0.0067, smaller by a factor of about forty. The $\sqrt{3}$ comes from the factor $1 - y^2$ at the escape threshold, not from the threshold itself.

3.4 The two criticality measures

With $\omega_k = \tau_k$ the radius near the saddle is about $\bar{\mu}\varepsilon$, a constant rather than a quantity proportional to $|y_k|$, so growth becomes arithmetic. The ratio of the two escape times is measured at 8.64, and it is stable across the grid: the individual cells give 8.635, 8.644, 8.644 and 9.571, with no trend in ε or in $\bar{\mu}$. The structural prediction, that the ratio does not depend on ε , therefore holds.

The predicted value $2\ln(1/(2y_0)) = 12.43$ does not. Repeating (6) for $\tau = \varepsilon(1 - 3y^2)$ gives

$$\int_{y_0}^{1/2} \frac{dy}{1 - 3y^2} = \frac{\text{artanh}(\sqrt{3}/2)}{\sqrt{3}} + \mathcal{O}(y_0) = 0.7604,$$

so the ratio should be $6.3585/0.7604 = 8.362$, within three per cent of what is measured. The arithmetic prediction $k_{\text{esc}} \approx 1/(2\bar{\mu}\varepsilon)$ omits the decrease of τ_k as $|y_k|$ grows, which is what the factor 0.7604 against 0.5 records.

The extra curvature is not free. At $\bar{\mu} = 1$ and $\varepsilon = 10^{-2}$ the $\|g\|$ -anchored run uses 1502 Hessian-vector products and the τ -anchored run 940, so on that cell the second measure is cheaper in total because it escapes sooner, while costing more per iteration.

3.5 Three outcomes

At $\text{tol} = 10^{-5}$, 136 of the 160 R-grad and R-DFO cells end in false convergence, against 12 fast escapes, 11 slow and 1 that never escapes. Rerunning those 136 cells with the second-order

stopping test at $\text{tol}_H = 10^{-6}$ leaves none of them false-convergent: 68 become fast escapes, 29 slow, and 39 do not escape within the budget. A method that refuses to stop at the saddle escapes slowly instead, and in 39 cells the wrong answer becomes one that does not finish.

The radius measure and the stopping test are two different uses of second-order information and are reported separately throughout.

3.6 The R-DFO arm

Started at $\Delta_0 = \zeta \|g_0\|$, so that the entry condition holds at $k_0 = 0$, the R-DFO runs stall exactly as the proposition requires. The measured θ_k settles at ζ rather than at $\gamma_3\zeta$, and the fitted rate is correspondingly about half the bound: at $\zeta\varepsilon = 10^{-3}$ the fitted rate is 0.00103 against the bound $\ln(1 + \gamma_3\zeta\varepsilon) = 0.00200$. The bound of Proposition 4 therefore holds and is not tight, the factor γ_3 being the worst case of an expansion that the next iteration undoes. Scaling by ζ rather than by $\gamma_3\zeta$ puts the collapse at 6.048, near the R-grad value.

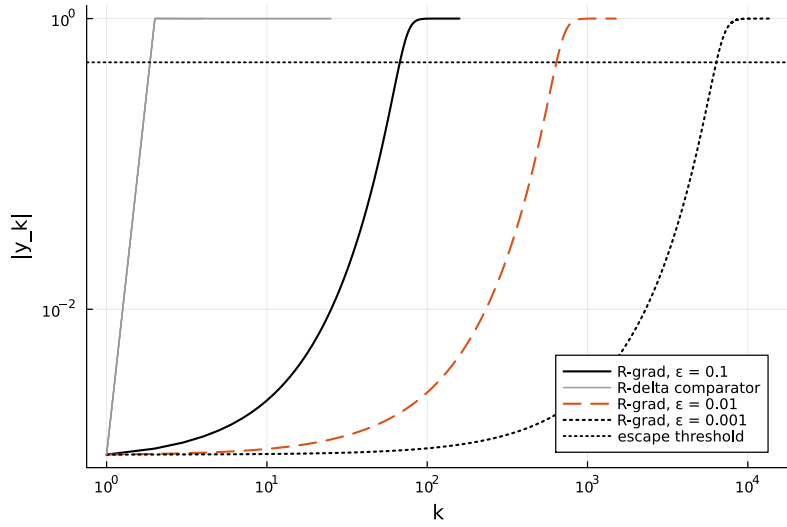


Figure 1: $|y_k|$ against k for three values of ε at $\bar{\mu} = 1$, with the R-delta comparator in grey and the escape threshold dotted.

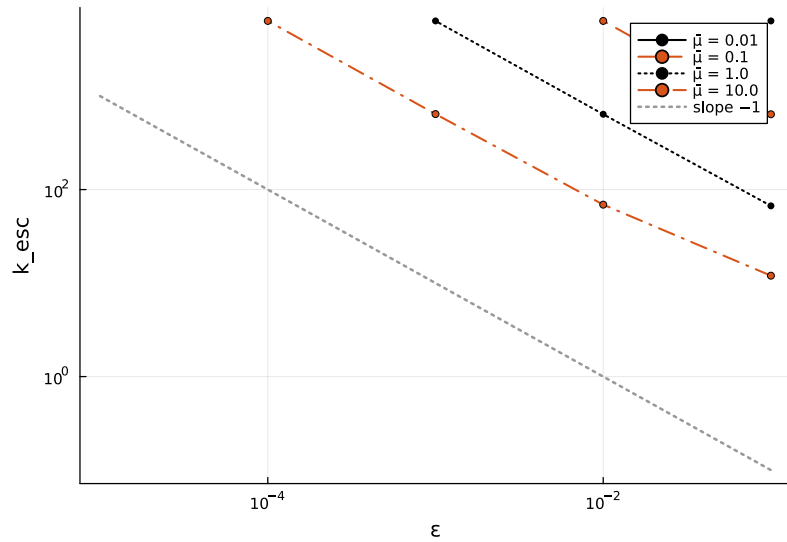


Figure 2: k_{esc} against ε , one series per $\bar{\mu}$, with the predicted slope -1 .

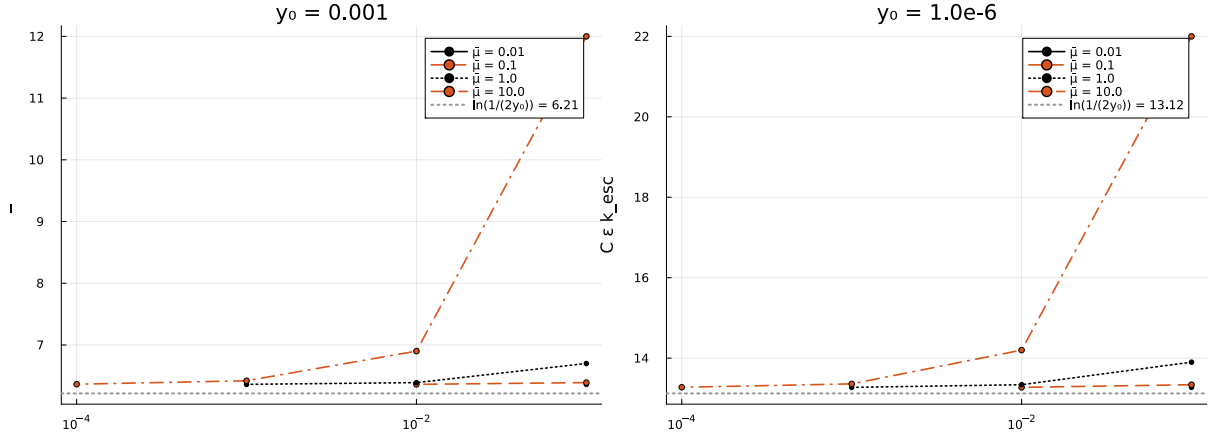


Figure 3: The collapse. $\bar{\mu}\varepsilon k_{\text{esc}}$ against ε for the whole grid, one panel per y_0 , with $\ln(1/(2y_0))$ marked.

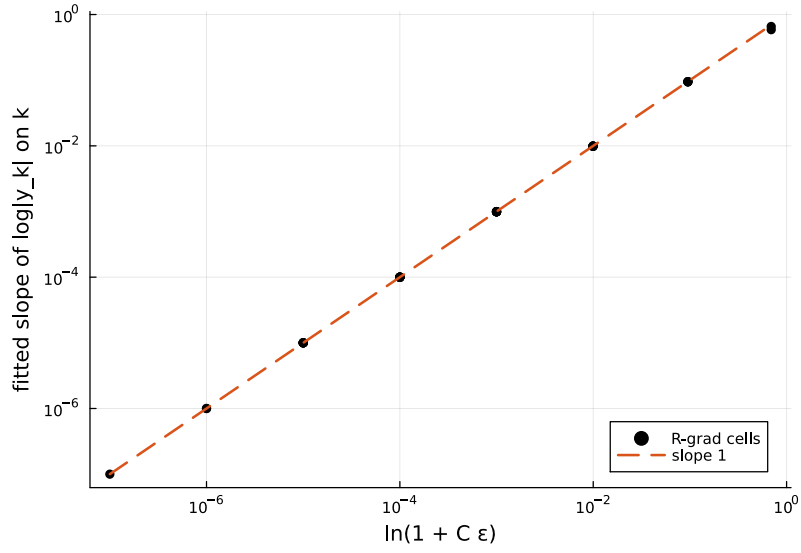


Figure 4: The fitted geometric rate against $\ln(1 + \bar{\mu}\varepsilon)$, with the line of slope 1.

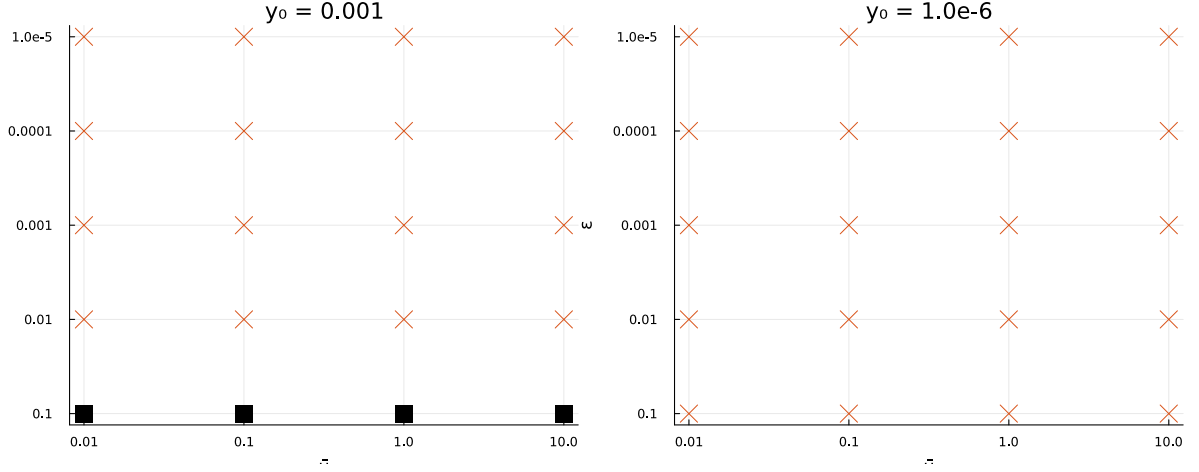


Figure 5: The three outcomes over the $(\bar{\mu}, \varepsilon)$ grid at first-order tolerance 10^{-5} , which is the tolerance that draws the boundary. Crosses mark false convergence, circles fast escape, squares slow escape or none.

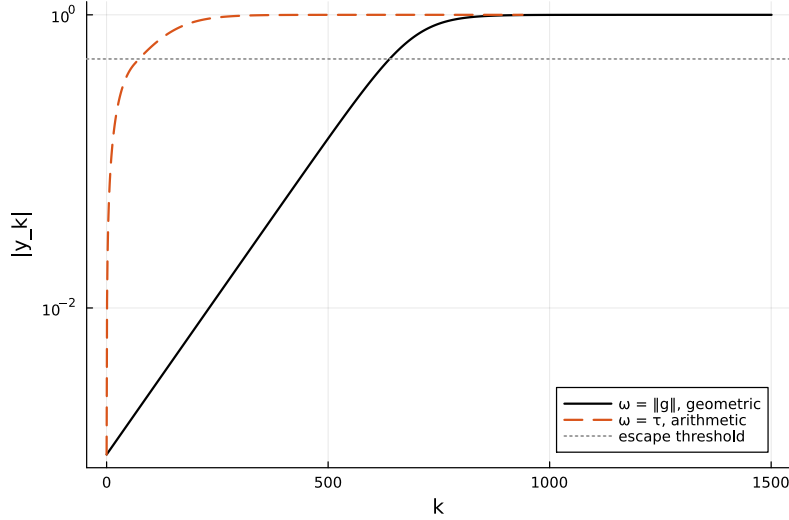


Figure 6: $\omega_k = \|g_k\|$ against $\omega_k = \tau_k$ at $\bar{\mu} = 1$, $\varepsilon = 10^{-2}$, $y_0 = 10^{-3}$. Geometric growth against arithmetic growth.

4 Findings for the existing subsection

These are proposals. We did not edit the source. Line numbers are from `_Thesis.FINAL/Article3/Survey-part`

1. **Line 1109, a self-reference.** The caption ends `This table isolates the model axis; Section~\ref{subsec: flat} adds the radius axis.` The table sits inside the subsection carrying `\label{subsec: flat}` at line 1051, so the cross-reference points at its own section.
2. **Line 1096, a false statement.** The source reads $\|g_0\| \approx \varepsilon y_0 = 10^{-10}$. With $x_0 = (0.8, y_0)$ the measured $\|g_0\|$ is 0.800000, larger by a factor of 8×10^9 . A sentence that says what happens without changing the claim: *the x component is solved first, after which $\|g_k\|$ falls to $\varepsilon |y_0| = 10^{-10}$ and the method meets its stopping criterion at a nonsingular saddle.* The number of iterations depends on Δ_0 , which the caption does not state. Under $\Delta_0 = 1$ the drop happens in one iteration.

3. **Line 1107, the invariant.** The caption says $\varepsilon k_{\text{esc}}$ is constant at $\approx \ln(1/y_0)$. Over the seven printed rows with a k_{esc} , the mean absolute residual is 0.4304 for $\ln(1/y_0)$, 0.3155 for $\ln(1/(2y_0))$ and 0.1716 for $\ln(1/(\sqrt{3}y_0))$. Restricted to the rows with $\varepsilon \leq 10^{-3}$, where the continuous limit applies, the residuals for $\ln(1/(\sqrt{3}y_0))$ are 0.0036, 0.0068 and 0.0004. The $\frac{1}{2}$ improves the fit, and $\sqrt{3}$ improves it further, for the reason given at (6).
4. **Line 1125, an em dash.** The last row contains a --- cell, which sets an em dash. Leave it empty or mark it n/a.
5. **The status column.** **stagnated** and **converged** record two different events. **converged** means $\|g_k\| \leq 10^{-9}$, a mathematical criterion. **stagnated** means the achieved reduction fell to the rounding level of f with a numerically nil step, an arithmetic event that says the run stopped making progress rather than that it found anything. The column should say which.

5 Where this belongs

Proposition 2 and Corollary 3 are candidates for the subsection of Part II carrying **subsec: Danger of saddle points**, immediately after **cor: convergence criticality anchored saddle**, which they quantify. The experimental sections are candidates for the subsection of the numerical study carrying **subsec: flat**, whose own note asks for the $(\bar{\mu}, \varepsilon)$ grid with the R-delta comparator.